
Question #: 1

A 29-year-old man comes to the emergency department because of a 2-day history of pain and swelling in his right knee after a fall. Physical examination shows redness and swelling in the skin over the knee. The patella could not be palpated through the swelling, and a diagnosis of bursitis is made. In which of the following structures is the inflammation most likely located?

A. Infrapatellar fat pad
✓B. Prepatellar bursa
C. Deep infrapatellar bursa
D. Subcutaneous infrapatellar bursa
E. Suprapatellar bursa

Rationale: The prepatellar bursa is located anterior to the patella in the subcutaneous tissue between the skin and the patella. If inflamed, it causes pain and swelling over the knee joint.

Question #: 2

A 20-year-old man is brought to the emergency department by paramedics 1-hour after he sustained a knife wound to the right side of his groin. The wound appears to be in the femoral triangle, and there is concern that the nerve may have been damaged. Which of the following movements will most likely be lost if the nerve in this area is injured?

A. Adduction of thigh
B. Extension of thigh
✓C. Extension of the knee
D. Lateral rotation of hip
E. Flexion of knee

Rationale: The nerve located in the femoral triangle is the femoral nerve. It innervates the muscles of the anterior compartment of the thigh, including the quadriceps muscles which extend the knee joint. Therefore if this nerve is damaged, the patient will not be able to extend the knee joint.

Question #: 3

A 17-year-old boy comes to the physician for a routine examination. Physical examination shows that stroking the medial side of the thigh results in a reflex that pulls up the cremaster. Which of the following nerves is most likely responsible for the efferent pathway of this reflex??

A. Ilioinguinal nerve
✓B. Genital branch of genitofemoral nerve
C. Obturator nerve
D. Saphenous nerve
E. Iliohypogastric nerve

Rationale: Ilioinguinal nerve (is a sensory nerve and thus afferent pathway)
Genital branch of genitofemoral nerve (correct: this is a motor branch to cremaster muscle and thus an efferent pathway of the cremaster reflex)
Obturator nerve (is medially located but is not related to the cremaster reflex)
Saphenous nerve (is deep to the fascia lata in the thigh)
Iliohypogastric nerve (is located above the inguinal ligament and not related to cremaster reflex)

Question #: 4

A 74-year-old woman is brought to the physician by her son for a routine examination. Physical examination shows strong peripheral pulses including the pulse behind the medial malleolus. Which of the following arteries is most likely palpated in this area?

A. Anterior tibial
B. Medial plantar
C. Fibular
D. Dorsalis pedis
✓E. Posterior tibial

Rationale: The posterior tibial artery runs behind the medial malleolus as it passes through the tarsal tunnel. It then divides in lateral and medial plantar arteries.

Question #: 5

A 34-year-old male was brought to the emergency department after he was involved in a motor vehicle collision that left him with a severely injured lower left limb. He is stabilized, blood samples taken for testing and cross match and immediately taken to the operating theater. Imaging studies show multiple fractures of the distal leg. A below-knee amputation was done. Which of the following cut structures belong to the lateral compartment of this region?

A. Superficial fibular nerve, fibularis longus and tertius

B. Deep fibular nerve, tibialis anterior and posterior

✓C. Superficial fibular nerve, fibularis longus and brevis

D.
Tibial nerve, flexor digitorum and flexor hallucis longus

E. Common fibular nerve, fibularis longus, brevis and tertius

Rationale: The lateral compartment of the leg is innervated by the superficial fibular nerve and comprises muscles responsible for eversion of the foot, fibularis longus and brevis. The fibularis tertius is an extension of the extensor digitorum longus a muscle belonging to the anterior compartment.

Question #: 6

A 59-year-old man comes to the emergency department because of a 2-hour history of a fast, irregular heartbeat. He appears diaphoretic and in respiratory distress. Physical examination shows tachycardia. In which of the following structures is the most likely location of the preganglionic cell bodies of the nerve fibers responsible for this patient's sweating?

A. Lateral horn S2-S4
B. Paravertebral ganglion
C. Preaortic ganglion
✓D. Lateral horn T1-L2

Rationale: Sweating and increased heart rate are responses generated by the sympathetic system. The preganglionic cell bodies of sympathetic system is located in the lateral horn.

Question #: 7

A 25-year-old woman comes to the physician because of a 1-week history of numbness (paresthesia) of the skin in the suprapubic region of the abdomen. Two weeks ago, She underwent an emergency appendectomy. Which of the following nerves is most likely injured?

A. Genitofemoral
✓B. Iliohypogastric
C. Subcostal
D. Spinal nerve T10
E. Ilioinguinal

Rationale: Iliohypogastric and Ilioinguinal nerve get damaged during open appendectomy, . Iliohypogastric nerve is a sensory nerve that provides lateral and anterior cutaneous branches supplying the posterolateral gluteal skin and skin in the pubic region.

Question #: 8

A male newborn is examined for an abnormality in his right hand as shown in the picture. Which of the following is the most likely embryological explanation for this abnormality?

A. Disruption of the anterior-posterior pattern
B. Failure of development of the notches between the digital rays
C. Suppression of limb bud development in the 4th week
✓D. Failure of development of one or more digital rays
E. Loss of apoptosis of the webbing between the digits

Rationale: Development of the digits begin with the fragmentation of apical ectodermal ridge and digit rays form. Apical ectodermal ridges at the tips of each digit, induces the mesenchyme to condense and transform into the phalanges.

Attachment:

for question 215082.jpg

Question #: 9

An 80-year-old woman is brought to the emergency department by ambulance after being found collapsed on her bathroom floor. Physical examination shows tenderness and limited movements in the hip joint. An x-ray is taken which is shown. Which of the following additional findings will most likely be present on physical examination of her leg?

A. Longer and externally rotated
B. Shorter and internally rotated
✓C. Shorter and externally rotated
D. Longer and externally rotated

Rationale: Shenton's line disruption: loss of contour between normally continuous line from medial edge of femoral neck and inferior edge of the superior pubic ramus. Neck of femur fractures are common injuries sustained by older patients.

Attachment:

for question 215079.jpg

Question #: 10

A 1-day-old male newborn has 'webbing' between the 2nd and 3rd toes of his left foot. Imaging studies show the phalanges of the two digits are separate and the webbing is skin only. Which of the following is the most likely diagnosis?

✓A. Syndactyly
B. Brachydactyly
C. Polydactyly
D. Ectrodactyly
E. Meromelia

Rationale: This patient has cutaneous syndactyly, where the webbing between the digits fails to degenerate. In osseous syndactyly there is a failure of the development of notches between the digital rays.

Question #: 11

A 17-year-old male comes to the physician because of 1-week history of right knee pain. Physical examination shows right knee joint effusion. He has discomfort with varus stress test to the knee. The anterior drawer test is negative. McMurray test is negative for pain and restriction. Radiological studies are unremarkable. Which of the following structures is most likely injured ?

✓A. Lateral collateral ligament injury
B. Medial meniscus injury
C. Lateral meniscus injury
D. Anterior cruciate ligament injury

Rationale: Varus stress test assesses lateral collateral ligament, therefore damage to the ligament can lead to pain and discomfort.

Question #: 12

A 20-year-old man comes to the physician because of a laceration to the posterior lateral wrist of his right hand. Physical examination shows loss of thumb extension but normal flexion. During physical examination, what other motor deficits will most likely be observed in this patient?

A. Loss of adduction of thumb
✓B. Weak abduction of thumb
C. Loss of adduction of digits 2-5
D. Loss of abduction of digits 2-5
E. Loss of opposition of thumb

Rationale: Abduction of thumb is possible by abductor pollicis brevis (innervated by recurrent branch of median Nerve) and abductor pollicis longus (innvervated by posterior interosseous nerve)

Question #: 13

A 6-year-old boy is brought to the physician by his mother because of a structural anomaly in his hand as shown in the photograph. Which of the following is the most likely embryological explanation for this anomaly?

A. Suppression of limb bud development in the 4th week
B. Failure of webs to degenerate between 2 or more digits
✓C. Disruption of the anteroposterior pattern
D. Failure of development of notches between the digital rays

Rationale: The patient has an extra digit on the medial side of the hand. This condition is called polydactyly and results from the disruption of the anteroposterior pattern of the limb.

Attachment:

Polydacyly 2_0.jpg

Question #: 14

A 50-year-old woman is admitted to the hospital for a modified radical mastectomy because of breast cancer. During recovery, she finds difficulty reaching for items on the kitchen shelf above shoulder level. Which of the following muscles most likely lost innervation due to this surgical intervention?

A. Levator scapulae
B. Pectoralis major
✓C. Serratus anterior
D. Subscapularis
E. Deltoid

Rationale: Correct: Serratus anterior is supplied by long thoracic nerve(C5, 6, 7), rotates scapula upward to allow lifting arm above 90 degrees and protracts the scapula to keep the medial border against the thoracic wall.

Incorrect: Subscapularis medial rotates the arm. Pectoralis major aids in Flexion, adduction and medial rotation of the arm.

Question #: 15

A 16-month-old girl is brought to the physician by her parents because of a 1-month history of bowing of the legs. Physical examination shows thickening of the wrists. Laboratory studies show elevated serum alkaline phosphatase (ALP) levels. Which of the following best describes the normal function of the vitamin most likely deficient in this patient?

✓A. Increases intestinal absorption of calcium and phosphate
B. Increases renal excretion of calcium and phosphate
C. Involved in hydroxylation of proline in collagen
D. Decreases synthesis of calbindins in the intestine
E. Decreases serum calcium and phosphate levels

Rationale: The child has vitamin D deficiency (rickets). 1,25-Dihydroxycholecalciferol increases the intestinal absorption of calcium and phosphate by inducing synthesis of calbindins (calcium binding proteins) in the intestinal mucosal cells. 1,25-DHCC increases serum calcium and phosphate levels by enhancing their intestinal absorption.

Question #: 16

A 60-year-old man comes to the physician because of a 2-month history of pain in his hips. He has a 4-year history of chronic renal failure and is on regular maintenance hemodialysis. Bone densitometry studies show decreased bone mineralization. Laboratory studies show low serum calcium and serum phosphate levels. Which of the following is most likely responsible for the findings in this patient?

A. Defective calcitriol receptor
B. Decreased activity of 25-hydroxylase
C. Decreased formation of vitamin D in the skin
✓D. Decreased activity of 1-hydroxylase
E. Decreased secretion of parathormone

Rationale: Patients with chronic renal failure have reduced activity of renal 1-hydroxylase. As a result, 1,25-DHCC levels are low resulting in less absorption of dietary calcium and phosphate. Patients with chronic renal failure are at risk of developing osteomalacia due to impaired activation of vitamin D and have reduced levels of 1,25-DHCC.

Patients with chronic liver disease have reduced activity of 25-hydroxylase.

Low serum calcium levels in patients with chronic renal failure will stimulate secretion of parathormone and this results in secondary hyperparathyroidism.

Patients with reduced sunlight exposure have impaired formation of vitamin D in the skin.

Question #: 17

A 3-year-old boy is brought to the physician because of a 3-month history of bowing of legs. Laboratory studies show low serum calcium and phosphate levels; elevated levels of 25-hydroxycholecalciferol and 1,25-dihydroxycholecalciferol (DHCC). A diagnosis of vitamin D resistant rickets is made. Which of the following is most likely abnormal in this child?

A. 1-hydroxylase activity in kidney
B. 25-hydroxylase activity in liver
✓C. Calcitriol receptor in intestine
D. Calcitonin secretion in response to hypercalcemia
E. Calcium sensor in parathyroid gland

Rationale: Vitamin D resistant rickets is an inherited disorder due to a mutation in the calcitriol (1,25-DHCC) receptor in intestine. The calcitriol binds to its receptor which is intracellular and the hormone-receptor complex binds to DNA and alters gene expression. It increases expression of the calbindins (calcium binding proteins) in the intestinal mucosal cell which results in increased dietary calcium and phosphate absorption. This in turn increases serum calcium levels.

Question #: 18

A 35-year-old woman comes to the physician because of a 2-month history of bone pain and weakness. Her dietary history shows megadoses of vitamin D supplementation. Which of the following laboratory findings is most likely in this patient?

✓A. Increased serum calcium levels
B. Decreased serum 25-hydroxy vitamin D levels
C. Decreased serum phosphate levels
D. Decreased serum calcium levels
E. Increased serum PTH levels

Rationale: Hypervitaminosis D is characterized by increased serum calcium and phosphate levels due to excessive dietary calcium absorption. These patients are at risk for developing calcification of soft tissues like kidney.
25-hydroxy D levels are elevated in hypervitaminosis D. PTH levels are low due to high serum calcium levels which suppress PTH secretion.

Question #: 19

Two siblings aged 3 and 5 years are brought to the physician by their parents because of a 4-week history of awkward gait. Physical examination shows the findings in the photograph. A diagnosis of a nutritional deficiency disorder is made. Which of the following is the strongest predisposing risk factor for the development of this disorder?

A. Decreased dietary fiber content
✓B. Decreased exposure to sunlight
C. Essential fatty acid deficiency
D. Decreased vitamin C intake
E. Decreased dietary protein content
F. Decreased dietary caloric content

Rationale: The children have bowing of the legs which is most commonly due to vitamin D deficiency (Rickets). Decreased exposure to sunlight as during the winter months in the temperate climates is a common risk factor for rickets.
Decreased dietary fiber content increases risk of constipation.
Essential fatty acid deficiency results in skin changes.
Vitamin C deficiency manifests as scurvy - bleeding gums and decreased stability of collagen.
Marasmus is due to reduced caloric content and protein in diet. Characterized by extensive loss of adipose tissue and thin arms and legs. Kwashiorkor is characterized by loss of body weight, skin changes and edema.
Attachment:

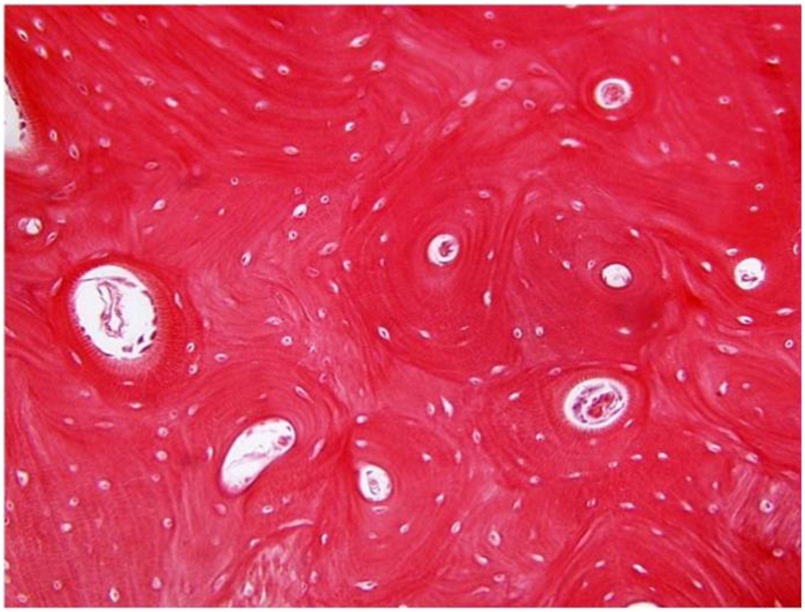
58473.JPG

Question #: 20

A 68-year-old man comes to the physician because of a 3-month history of pain in the right lower limb. A biopsy is collected and shown in the attached photomicrograph. Which of the following is a characteristic feature of this tissue?

A. Matrix contains abundant type II collagen
B. Matrix contains abundant hyaluronic acid
C. Cells are organized as isogenous groups
D. Matrix contains abundant elastic fibers
✓E. Matrix contains abundant type I collagen

Rationale: The major structural component or fiber of bone's extracellular matrix is type I collagen.
Attachment:



Picture38.JPG

Question #: 21

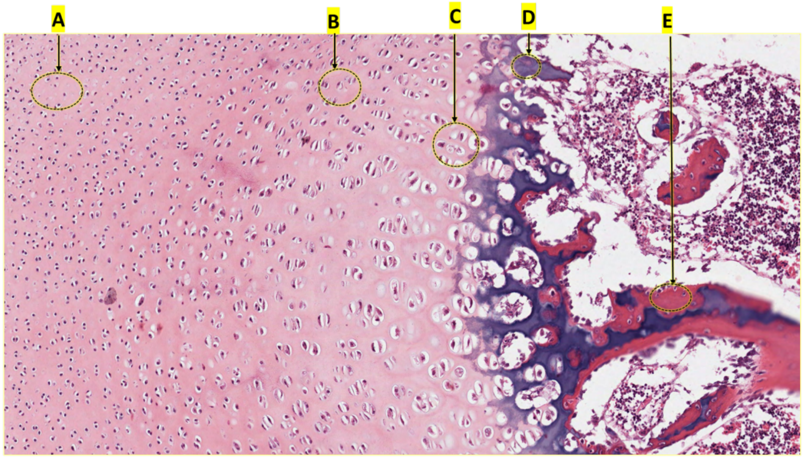
A new chemotherapy drug is shown inhibit vertical growth of the long bones. Animal models show that the drug is selectively toxic to osteoblast at the epiphyseal plate. Which of the following sites most likely contains cells that will be affected by the drug?

A. A
B. B
C. C
D. D
✓E. E

Rationale: The correct answer is E - the zone of resorption where calcified cartilage. In this zone calcified cartilage is broken down and osteoprogenitor cells from the bone marrow invades, differentiate into osteoblasts and begin to lay down bone.
A is the resting zone or of reserve cartilage which contains the stem cells of the epiphyseal plate .
B = proliferative (note the stacks of chondrocytes) and D is the zone of calcified cartilage indicated by

the deep basophilia of the matrix.

Attachment:



3 Epiphysis.png

Question #: 22

A 43-year-old man comes to the physician because of 3 month history of difficulty in swallowing. He has smoked cigarettes for more than 20 years. Physical examination of the oral cavity shows a large mass on the epiglottis. Which of the following types of cartilage are most likely found in this region?

A. Hyaline cartilage
✓B. Elastic cartilage
C. Fibrocartilage
D. Fibrocartilage and hyaline cartilage
E. No cartilage is found here

Rationale: Elastic cartilage is distinguished by the presence of elastin in the cartilage matrix. It is found in the epiglottis of the larynx, external ear and auditory tube.

Question #: 23

A 3-year-old boy is brought to his physician by his parents because of a 2 hour history of a fall onto his arm. Physical examination shows swelling at the distal radius. Imaging studies show fracture of the distal radius with only discrete displacement of the fractured ends of the bone. Which of the following best describes the bony tissue located at the fracture site at this time?

A. Mainly type I and II collagen
B. Mainly proteoglycans and type II collagen
✓C. Blood clot formation
D. Type II collagen and hydroxyapatite

Rationale: This is a non displaced fracture the ends of the broken bone are close enough to begin healing. At this stage a clot or hematoma forms to prevent blood loss.
Later this will be replaced by granulation tissue (days after the fracture) . Granulation tissue is converted to fibrocartilage (type I and II collagen) which is later remodeled to bone (hydroxyapatite and type II collagen). Hyalin cartilage (type II collagen and proteoglycans is not involved in bone healing).

Question #: 24

A 53-year-old woman comes to the physician because of a 3-month history of sharp pain in the right buttock and thigh. Physical examination shows no abnormalities. MRI shows a herniation of the L4-L5 intervertebral disc. Which of the following is most likely characteristic feature of the cartilage component of this structure?

A. Matrix contains only Type II collagen fibrils
✓B. Matrix contains Type I and Type II collagen fibrils
C. Matrix contains abundant elastic fibers
D. Firmly attached perichondrium
E. High capability for repair

Rationale: Fibrocartilage consists of chondrocytes and their matrix in combination with dense connective tissue. It is typically present in intervertebral discs, the pubic symphysis and articular discs.

Question #: 25

A 5-year-old boy is brought to the physician by his mother because of a 3-month history of bowed legs. An x-ray shows kyphosis of the back, pigeon chest and bowing of the bones of both lower limbs characteristic of rickets. Which of the following best characterizes the cartilage that forms the foundation of the bones in the legs of this patient?

A. Predominantly type I collagen
B. Isogenous groups arranged in a linear pattern
✓C. Contains a high amount of GAGs
D. Highly vascularized
E. Lacks a perichondrium
F. Abundance of elastic fibers

G. Does not undergo appositional growth

Rationale: Calcium deficiency during growth in children causes rickets, a condition in which the bone matrix does not calcify normally. The bones of the lower limbs are considered to be long bones. These usually begin on a foundation (model) of hyaline cartilage (endochondral ossification). Hyaline cartilage matrix is characterized by an abundance of GAGs, some sulphated, with high affinity for water molecules (imparts resilience, allows difusion)

A- fibrocartilage
B - fibrocartilage
D - cartilage is avascular
E - feature of fibrocartilage
F - elastic cartilage
G - fibrocartilage (that lacks a perichondrium)

Question #: 26

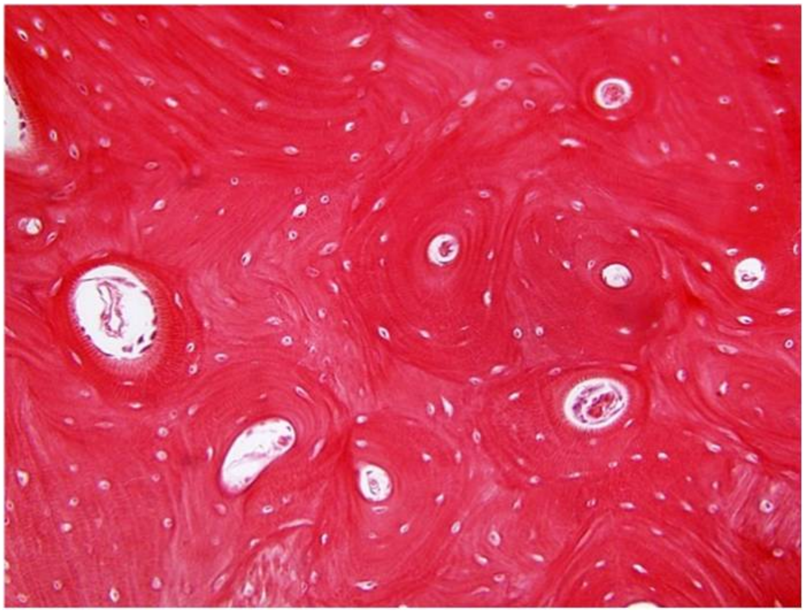
A 2-year-old girl is brought to the physician by her parents because of a 1-week history of pain in the wrist. Physical examination shows a swollen left wrist. There are no obvious deformities. An X-ray of the wrist shows no signs of fracture and the well-defined radiolucent line on the distal radius remains intact. Which of the following is most likely a feature of this region of the bone?

A. Contains longitudinal vascular canals
✓B. Composed of five histological zones
C. Contains transverse vascular canals
D. Promotes appositional growth
E. Consists of Haversian systems

Rationale: The radiolucent line referred to above is that of the epiphyseal growth plate. This region separates the primary ossification centre, which originally began in the diaphysis of long bones from the secondary ossification center, which happens at the epiphysis. Histologically, it is made up of 5 zones.

A - Haversian canal found in compact bone
C - Volkmann's canal - found in compact bone
D - the epiphyseal growth plate promotes growth in length (interstitial), not growth in width (appositional growth)
E - a feature of mature compact bone

Attachment:



Picture38.JPG

Question #: 27

A 75-year-old woman is brought to the emergency department by her husband because of severe pain and a left hip deformity after falling out of bed 4-hours ago. Physical examination shows tenderness and an externally rotated left lower limb. An x-ray of the pelvis and hip shows a fracture in the neck of the left femur. Bone mineral density measurement shows severe osteoporosis. Which of the following best characterizes the cartilage most likely to be affected in this disease process?

A. Imparts tensile strength to tissues
B. Preponderance of elastic fibers
✓C. Rounded chrondrocytes in the intermediate zone
D. Hypertrophied chondrocytes in the zone of hypertrophy
E. Found in areas of high stress

Rationale: Osteoporosis is characterized by the progressive loss of bone density caused by increased activity of osteoclasts and bone resorption. There is also marked destruction of the articular cartilage, which sits over the bone and in the joint spaces. This cartilage also gets eroded and destroyed during the disease process. The articular cartilage is made up of 5 zones, and option C is one of them.

Question #: 28

A research group is developing a new drug to treat connective tissue disorders and wishes to investigate its effects on connective tissue elements in the body. The drug is give intravenously and x-rays are taken of different sites at various times. They discover that the drug has a predilection for bony tissue. Biopsy samples taken from some participants show destruction of specific connective tissue elements within the bone. Which of the following connective tissue fibers is most abundant in the affected tissue?

A. Type III collagen
B. Elastic
C. Type IV collagen
D. Type II collagen
✓E. Type I collagen

Rationale: Bone is specialized connective tissue, and as such is made up of cells and an extracellular matrix. That matrix is composed of fibers and ground substance. The predominant fiber in bone is type I collagen.

Type II is found mainly in cartilage and type iv in basement membrane while type III is more in areas of loose cells, forming a structural support meshwork such as in the immune system.

Question #: 29

A group of researchers is studying the effects of growth hormone on height. The study involves observation of the epiphyseal growth plate in vitro. Which of the following best characterizes the zone most likely influenced by the hormone being studied?

A. Inactive chondrocytes
B. Calcified inter-lacunae matrix
C. Greatly enlarged chondrocytes
✓D. Chondrocytes undergoing rapid mitotic division
E. Resorption of calcified bone-cartilage complex

Rationale: The chondrocytes found within the zone of proliferation in the epiphyseal growth plate possess receptors for growth hormone. Under the influence of growth hormone, these cells would promote longitudinal growth of the bone. This zone is characterized by rapid mitotic division.

A - zone of reserved cartilage
B - zone of calcified cartilage
C - zone of hypertrophy
E - zone of resorption

Question #: 30

A 45-year-old woman comes to the physician because of a 6-month history of knee pain and stiffness. Her vital signs are within normal limits. An MRI of her knee shows thinning of the articular cartilage and early signs of osteoarthritic degeneration. Examination of a biopsy specimen of the articular cartilage shows cells that appear in isogenous groups within lacunae, surrounded by a distinct pericellular matrix, then a more fibrillar territorial matrix, and finally a more homogeneous interterritorial extracellular matrix. Which of the following cells is most likely being observed?

✓A. Chondrocytes
B. Chondrogenic cells
C. Chondroclasts
D. Fibroblasts
E. Chondroblasts

Rationale: Cells of cartilage include all options. Fibroblasts are usually found in fibrocartilage and in the outer layer of the perichondrium. Chondroblasts derive from chondrogenic cells, and are never found in isogenous groups or lacunae. Their major function is to secrete the matrix. When chondroblasts are found in lacunae, they are now called chondrocytes. Based on the description given, the cells observed are chondrocytes. Chondrocytes also produce some matrix, and in so doing, help to maintain the already produced matrix. Chondrogenic cells are found in the inner layer of the perichondrium, and are never surrounded by matrix. Chondroclasts are derived from monocytes and play a role in cartilage remodeling.